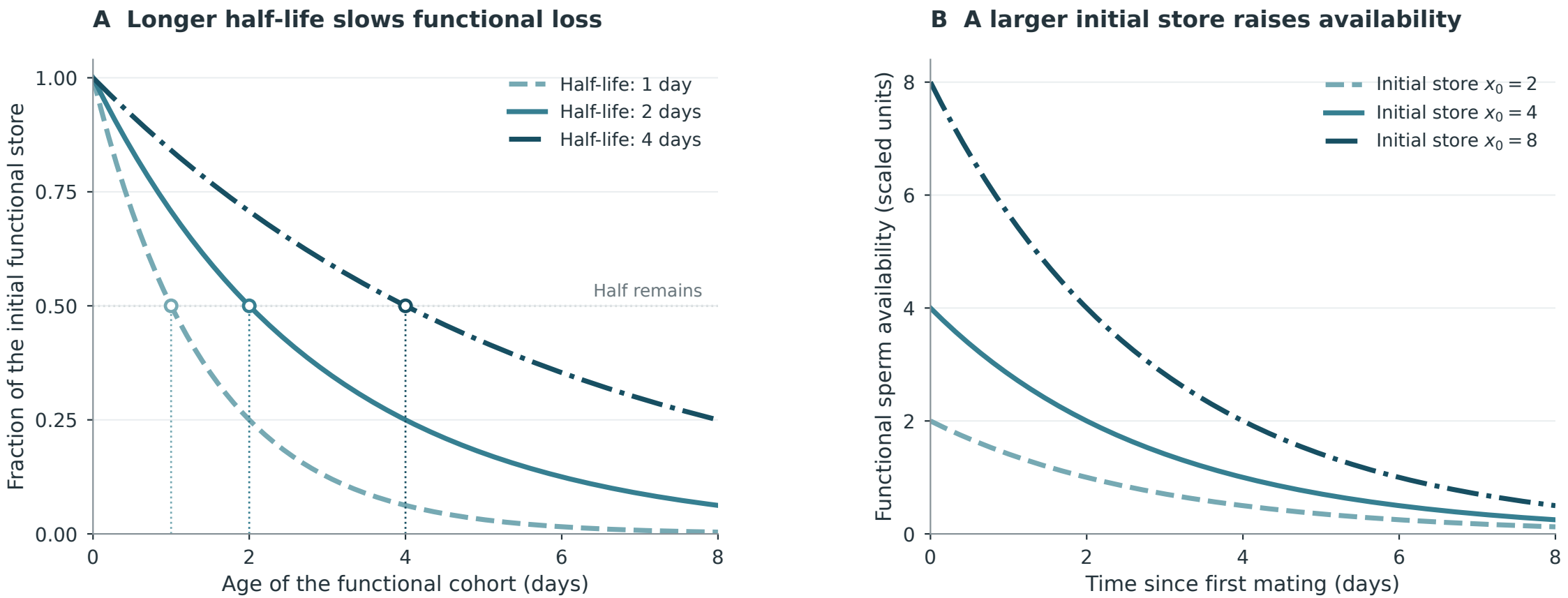
1 Functional lifespan and initial amount are different traits

A larger starting store can preserve availability without changing the rate of functional loss.

$$S(t) = 2^{-t/h} \quad ; \quad x(t) = x_0 S(t)$$



Illustrative model. A: equal initial cohorts. B: a common functional half-life of 2 days.

Argument: measure lifespan and initial amount separately.

Panel A holds initial availability constant and varies the functional half-life. The open circles mark the age at which half of each initial cohort remains functional and available. Panel B holds the half-life at two days and varies the initial store. Increasing the starting amount raises the entire availability curve without changing proportional decay. In this model, producing more functional sperm and preserving existing sperm are distinct ways to maintain availability. A larger store is a possible consequence of greater production, not a direct measurement of production.

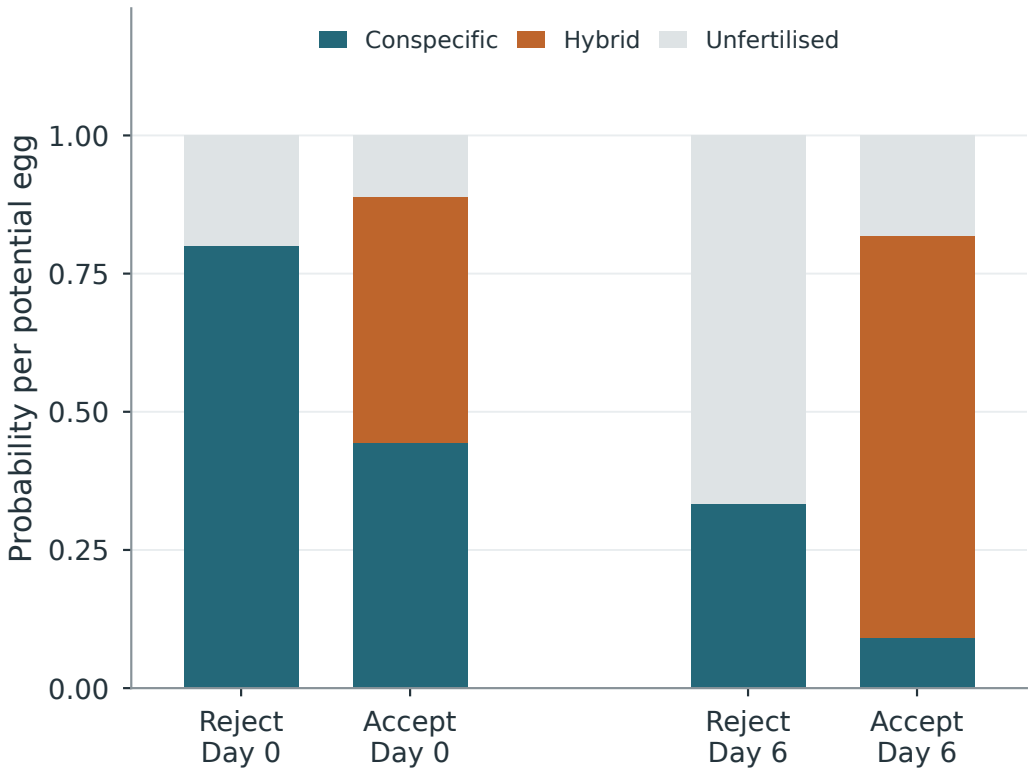
Measurement: S(t) is the surviving functional fraction of the original store, not the fraction alive among sperm retained at a later sample. Initial transfer, entry into storage, death, loss of function, use and ejection require separate measurements. These trajectories do not identify the biological cause of loss.

2 Remating can help after the conspecific store declines

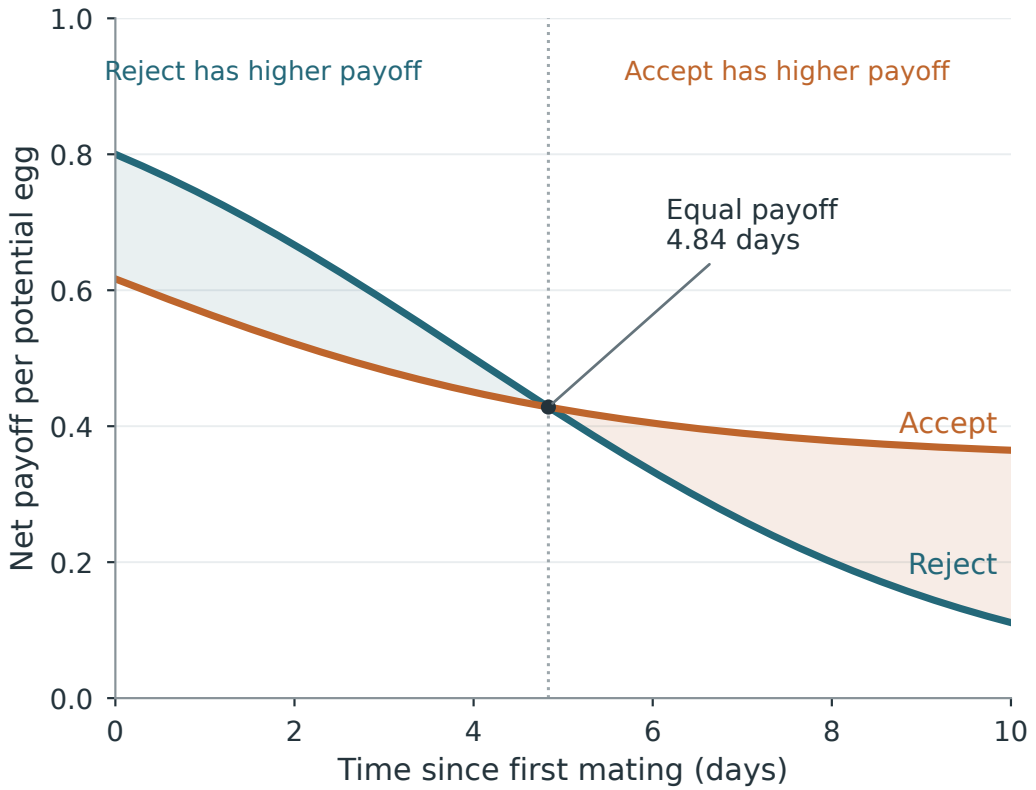
New sperm can reduce fertilisation failure while also replacing conspecific fertilisations with hybrid fertilisations.

$$W_{\text{reject}} = \frac{x}{1+x} \quad ; \quad W_{\text{accept}} = \frac{x+wy}{1+x+y} - c$$

A Fertilisation outcomes before fitness weighting



B Net payoff after offspring fitness and cost



Illustrative values: $x_0 = 4$, $h_C = 2$ days, $y = 4$, $w = 0.5$, $c = 0.05$. Sperm amounts are scaled units.

Argument: fertilisation assurance and offspring fitness can favour different actions.

Panel A partitions potential eggs into conspecific fertilisations, hybrid fertilisations and fertilisation failures. At either age, accepting the new ejaculate reduces fertilisation failure but also reduces the probability of conspecific fertilisation. Panel B weights conspecific offspring by 1 and hybrid offspring by 0.5, then subtracts a mating cost of 0.05 from acceptance. Rejection gives the higher payoff early; acceptance gives the higher payoff after 4.84 days. The original conspecific sperm remain available after remating. The change in the preferred action is caused by declining conspecific availability, not a change in hybrid fitness.

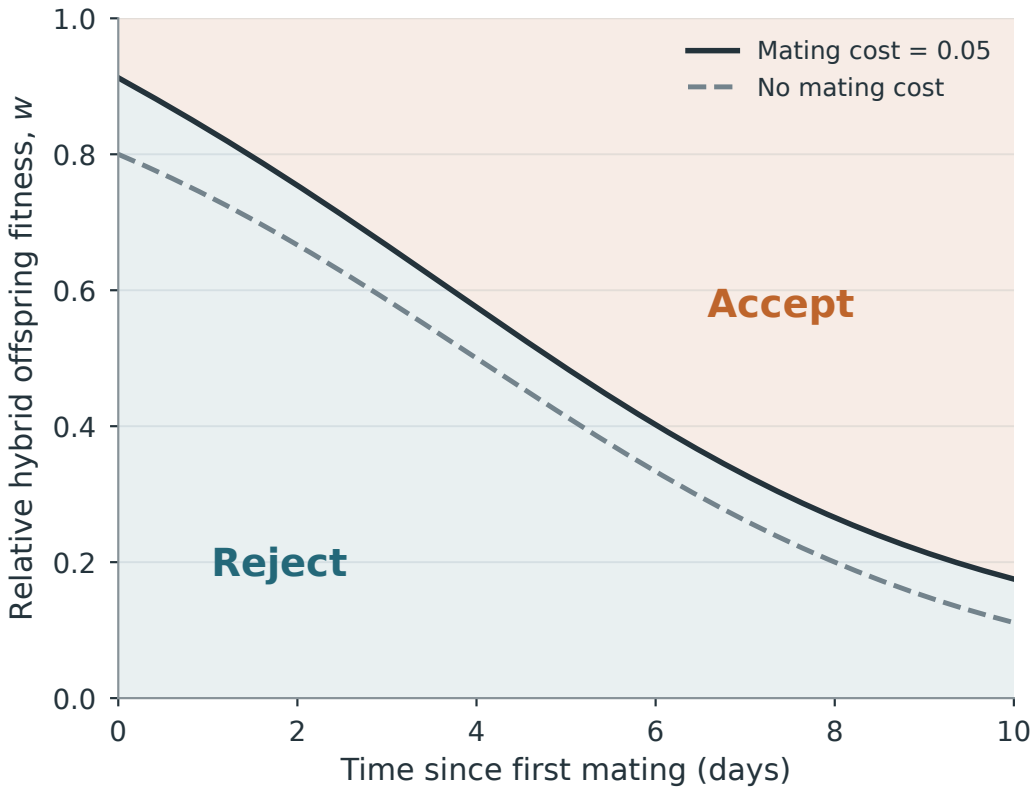
Scope: payoffs describe the next short reproductive bout, with a fixed pool of potential eggs and no further mating opportunity. Cost is an absolute payoff deduction, not a 5% survival reduction. The comparison assumes access to the relevant state information; it does not demonstrate that females assess sperm condition.

3 Hybrid fitness and sperm longevity shift the acceptance threshold

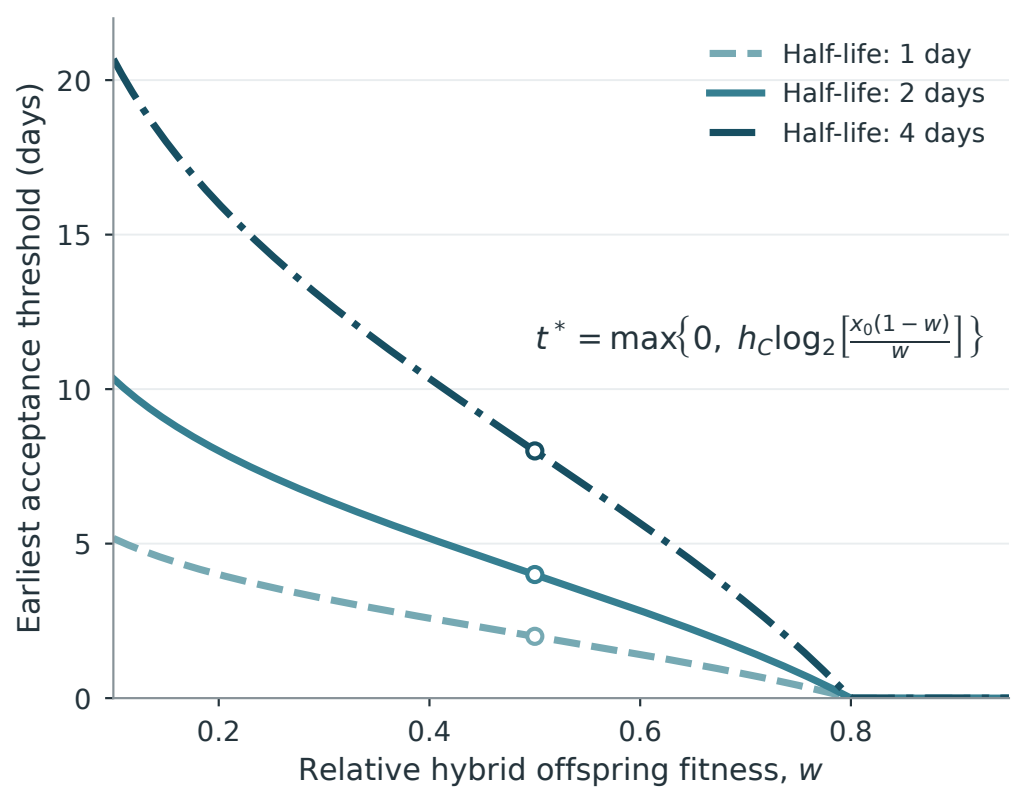
Acceptance has the higher payoff above the boundary. Longer persistence delays the cost-free threshold.

$$w_{\text{crit}}(t) = \frac{x(t)}{1 + x(t)} + \frac{c\,[1 + x(t) + y]}{y}$$

A The boundary shifts when mating is costly



B Longer half-life delays the cost-free threshold



A: $x_0 = 4$, $h_C = 2$ days, $y = 4$. B: $x_0 = 4$, $c = 0$, $y > 0$. Values are illustrative.

Argument: lower hybrid fitness and longer sperm persistence delay beneficial heterospecific mating.

Panel A rearranges the payoff inequality into a critical hybrid fitness. The solid boundary includes mating cost 0.05; the dashed boundary sets the cost to zero. Shaded regions refer to the solid boundary. At a fixed age, acceptance has the higher payoff above that boundary. Lower hybrid fitness requires a more depleted conspecific store before acceptance becomes beneficial. Panel B isolates the effect of conspecific half-life when mating is cost-free. Open circles show thresholds at hybrid fitness 0.5. The formula applies for 0 < w < 1 and a positive new ejaculate; negative calculated times are set to zero.

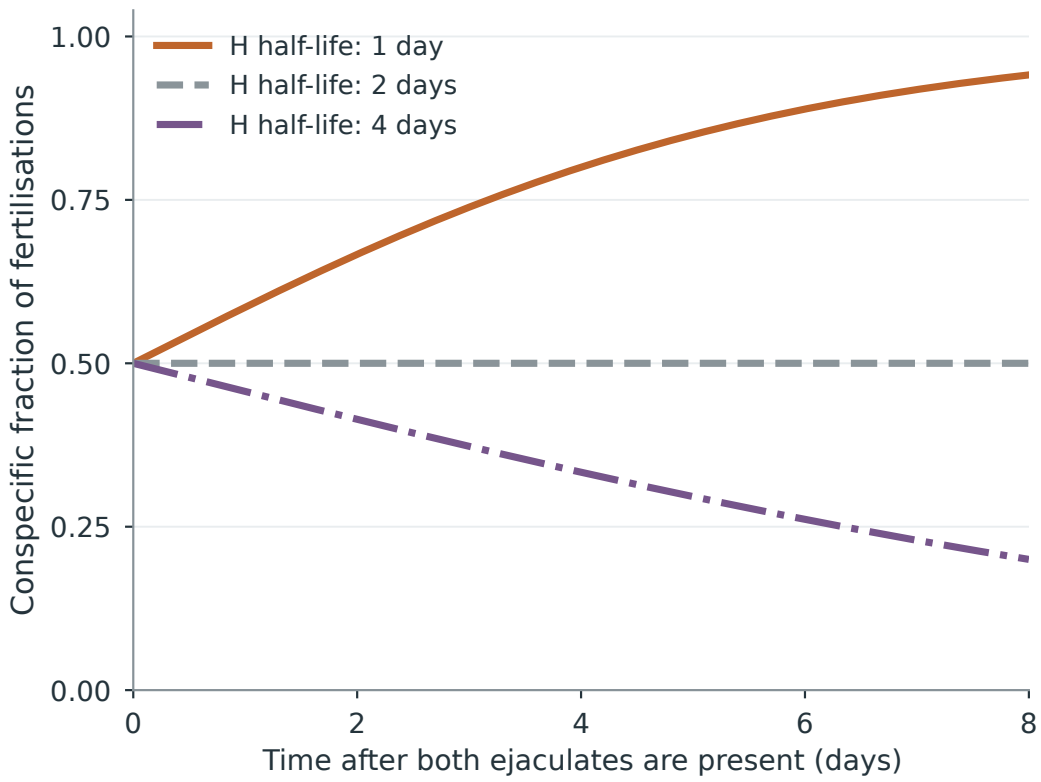
Inference: divergence predicts a later acceptance threshold only if it lowers hybrid fitness, holding the other model components fixed. The horizontal variable in B is hybrid fitness, not divergence time. Initial store and longevity have no specified evolutionary costs here, so this figure does not establish their evolutionary optima.

4 Conspecific share can rise while total fertilisation falls

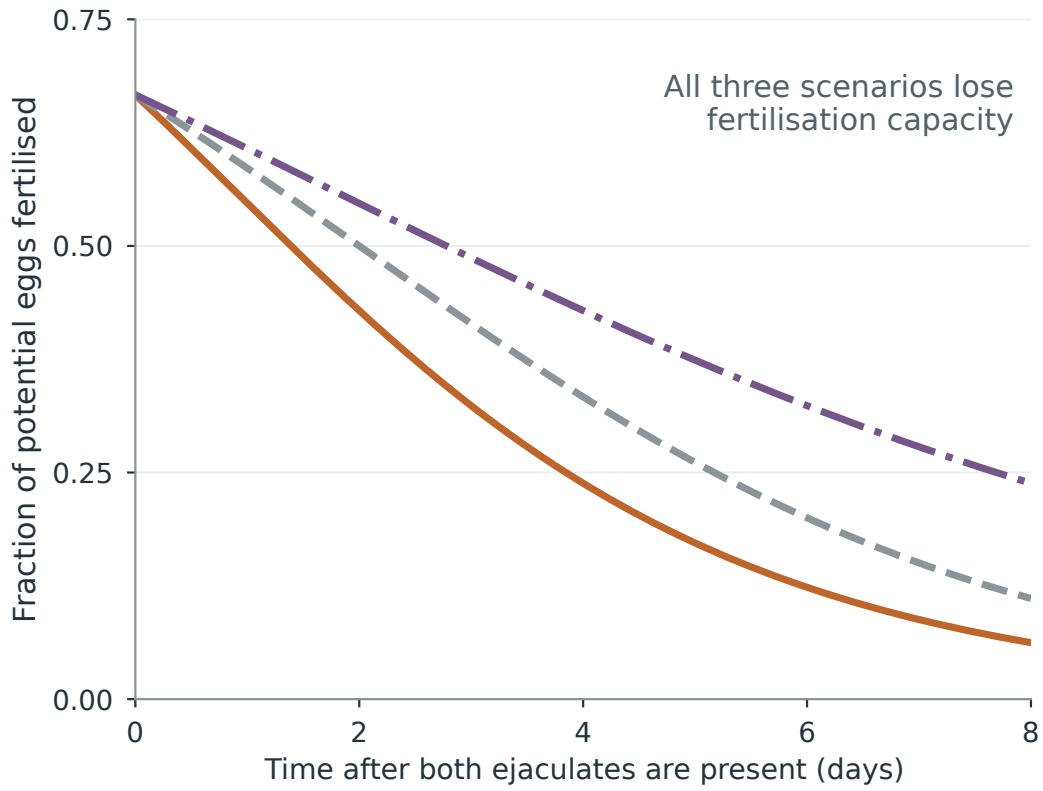
Both ejaculates start with equal functional availability. Conspecific half-life is fixed at 2 days.

$$X(u) = 2^{-u/h_C}, \quad Y(u) = 2^{-u/h_H} \quad ; \quad P_C = \frac{X}{X+Y}, \quad F = \frac{X+Y}{1+X+Y}$$

A Conspecific share among fertilised eggs



B Total fertilisation probability



Time zero is after both ejaculates are present. Matching colours and line styles identify the same scenario in both panels.

Argument: a rising conspecific fertilisation share does not imply increasing reproductive output.

Panel A shows conspecific share conditional on an egg being fertilised. The share increases when heterospecific sperm lose function faster, remains constant when half-lives are equal, and decreases when heterospecific sperm persist longer. Panel B shows the probability that a potential egg is fertilised at all. That probability declines in every scenario because both functional stores decline. For the one-day heterospecific half-life, conspecific share reaches about 94% at day eight, while total fertilisation falls to about 6%. Both quantities are required to interpret the biological consequence of a paternity bias.

Scope: equal initial stores isolate persistence and are a separate example from Figures 2 and 3. Half-lives are fixed; there is no new input, explicit sperm use or induced response to a competitor. These are snapshots across storage ages, not cumulative egg production. Differential persistence alone can generate this bias; it does not establish a competition-specific mechanism.